

A Quasi-Newton-like regularized approach to the order-value optimization problem*

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Abstract

The order-value function of order p returns, at each point, the p th smallest value among a finite family of functions $f_1, \dots, f_m$; minimizing it recovers the minimization of the minimum and of the maximum of the family in the extreme cases $p = 1$ and $p = m$, and provides a flexible tool for robust estimation, in which a prescribed number of outlying observations are discarded. We propose a regularized method for minimizing the order-value function subject to box constraints. At each iteration the method minimizes a strongly convex model that combines a piecewise-linear term built from the active gradients, a quasi-Newton curvature term defined by a symmetric positive semidefinite matrix, and a quadratic regularization; the curvature term carries second-order information without requiring explicit second derivatives, and a recently proposed first-order regularized method is recovered as the special case in which this term vanishes. We introduce an approximate optimality condition for the problem and prove that the method is well defined and attains the same worst-case iteration- and evaluation-complexity bounds as its first-order counterpart, so that the incorporation of curvature preserves the theoretical guarantees while offering the potential for higher-quality steps in practice. The applicability of the method to parameter-estimation problems with outliers is illustrated.

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1 Introduction

{sec:intro}

Many optimization problems involve objective functions that are obtained not by aggregating a collection of simpler functions, but by selecting one of them according to its rank. The prototypical example is the order-value function of order p , introduced in [3, 4]: given m functions $f_1, \dots, f_m$, its value at a point x is the p th smallest among $f_1(x), \dots, f_m(x)$. Functions of this kind belong to the broader class of generalized order-value functions—those whose value at x is governed by order relations among the elements of a set $\{f_i(x)\}_{i \in I}$ —a class systematized in [17]. Since the index that attains the p th position may switch abruptly as x moves, the order-value function is generally nonsmooth even when each f_i is continuously differentiable.

The extreme choices $p = 1$ and $p = m$ recover the minimum and the maximum of $\{f_i(x)\}$, respectively, so the order-value function interpolates between these two classical aggregates; the intermediate cases are the ones of practical interest. When x encodes portfolio positions and $f_i(x)$ is the loss incurred under scenario i , the order-value function coincides with the discrete value-at-risk (VaR) measure, a standard risk measure in financial risk management [16]. When $f_i(x)$ measures the misfit of a parametric model to the i th observation, minimizing the p th order-value function fits the parameters while discarding the $m - p$ worst-fitted observations. This makes order-value optimization a natural tool for robust estimation, where one seeks to remain insensitive to a minority of grossly corrupted data, or outliers.

The order-value optimization (OVO) problem was first formulated in [3], where a steepest-descent-type method was proposed and shown to converge to points satisfying a weak optimality condition. Sharper optimality conditions and a reformulation as a nonlinear program with equilibrium constraints were subsequently developed in [4]. A quasi-Newton method generalizing the scheme of [3] was introduced in [5], and a global optimization strategy combining multistart with a tunneling technique was proposed in [7]. On the computational-complexity side, [20] established that minimizing the order-value function over a polytope is NP-hard in the strong sense.

The systematic study of worst-case complexity in continuous optimization originates in the cubic-regularization analysis of [19], and complexity guarantees have since been obtained for a broad range of problems (see, e.g., [9, 10] and the monograph [13]). Within this line, [21] recently proposed a first-order method for box-constrained OVO that minimizes, at each iteration, a quadratically regularized piecewise-linear model of f , and established iteration- and evaluation-complexity bounds for reaching an approximate optimality condition.

In the present work we generalize the approach of [21] by enriching the regularized model with second-order information. At each iteration our method minimizes a model built from the same piecewise-linear convex term, a curvature term governed by a symmetric positive semidefinite matrix B , and a Tikhonov-type quadratic regularization. The matrix B is updated in a quasi-Newton fashion and therefore requires no explicit second derivatives, while the regularization renders the model strongly convex, so that each subproblem has a unique minimizer. Setting $B = 0$ recovers the first-order method of [21] exactly, so the proposed method is a strict generalization of it. We derive an approximate optimality condition for the box-constrained problem and prove that the method is well defined and attains the *same* worst-case iteration- and evaluation-complexity bounds as the first-order method. Thus the incorporation of curvature does not degrade the theoretical guarantees, while it offers the potential for higher-quality steps, and

71 hence faster convergence, in practice—a behavior we illustrate on parameter-estimation problems
 72 in the presence of outliers.

73 The remainder of the paper is organized as follows. Section 2 defines the box-constrained
 74 order-value optimization problem and the proposed regularized method. Section ?? introduces
 75 an appropriate optimality condition and establishes the well-definiteness, convergence, and com-
 76 plexity of the method. Section ?? reports illustrative numerical experiments on parameter esti-
 77 mation in the presence of outliers. Final remarks are given in Section ??.

78 *Notation.* The symbol $\|\cdot\|$ denotes the Euclidean norm. For $i = 1, \dots, n$, $e_i \in \mathbb{R}^n$ denotes the
 79 i th column of the

80 2 A quasi-Newton regularized method

81 Let $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, m$, be given and let $p \in \{1, \dots, m\}$ be fixed. The *order-value*
 82 *function of order p* is the mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$83 \quad f(x) = f_{i_p(x)}(x), \quad (1) \quad \{\text{ovo-function}\}$$

84 where $\{i_1(x), \dots, i_m(x)\}$ is a permutation of $\{1, \dots, m\}$ such that

$$85 \quad f_{i_1(x)}(x) \leq f_{i_2(x)}(x) \leq \dots \leq f_{i_m(x)}(x); \quad (2) \quad \{\text{fi-sort}\}$$

86 that is, $f(x)$ is the p th smallest value among $f_1(x), \dots, f_m(x)$. The boundary cases $p = 1$ and
 87 $p = m$ recover, respectively, the minimum and the maximum of $\{f_1(x), \dots, f_m(x)\}$.

88 If $f_1, \dots, f_m$ are continuous, then so is f . Differentiability, however, is not inherited: f may
 89 be nonsmooth at a point x at which the p th smallest value is attained by more than one index,
 90 i.e. whenever there exist $i \neq j$ with $f_i(x) = f_j(x) = f(x)$, since the active index $i_p(x)$ may then
 91 change as x varies.

92 We address the box-constrained order-value optimization problem

$$93 \quad \underset{x \in \Omega}{\text{Minimize}} \ f(x), \quad (3) \quad \{\text{ovo-problem}\}$$

94 where $\Omega := \{x \in \mathbb{R}^n \mid \ell \leq x \leq u\}$ with $\ell, u \in \mathbb{R}^n$ and $\ell_i < u_i$ for $i = 1, \dots, n$. Box con-
 95 straints arise naturally in parameter estimation, where the unknowns are confined to physically
 96 or statistically meaningful bounds.

97 For $\delta > 0$ and $x \in \Omega$, define the *active index set*

$$98 \quad I_\delta(x) := \{i \in \{1, \dots, m\} \mid f(x) - \delta \leq f_i(x) \leq f(x) + \delta\}, \quad (4) \quad \{\text{idelta}\}$$

99 the set of indices whose values lie within a band of half-width δ around $f(x)$. As $\delta \rightarrow 0$, $I_\delta(x)$
 100 shrinks to the set of indices that attain the order-value,

$$101 \quad I_0(x) := \{i \in \{1, \dots, m\} \mid f_i(x) = f(x)\}.$$

102 The method builds, around a reference point $\bar{x} \in \Omega$, a local model of f that combines the
 103 first-order information of the active components with curvature carried by a symmetric positive

semidefinite matrix $B \in \mathbb{R}^{n \times n}$. In a quasi-Newton framework, B is updated from successive gradients, so that no second derivatives are computed explicitly. For given $\bar{x} \in \Omega$, $B \succeq 0$, $\delta > 0$, and $\sigma > 0$, the regularized model is

$$\Psi(x; \bar{x}, B, \delta, \sigma) = \max_{i \in I_\delta(\bar{x})} \left\{ \nabla f_i(\bar{x})^T (x - \bar{x}) + \frac{1}{2} [(x - \bar{x})^T B (x - \bar{x}) + \sigma \|x - \bar{x}\|^2] \right\}. \quad (5) \quad \{\text{model}\}$$

The first term is the piecewise-linear convex function that captures the worst-case first-order behavior of f over the active components; the bracket adds the quasi-Newton curvature $(x - \bar{x})^T B (x - \bar{x})$ and a Tikhonov regularization $\sigma \|x - \bar{x}\|^2$. Since $\sigma > 0$, the model is strongly convex even when B is only positive semidefinite, so the subproblem (6) has a unique minimizer. For $B = 0$, the model reduces to the first-order regularized model of [21], which the present method thus generalizes.

Algorithm 2.1. Let $\delta > 0$, $\sigma_{\min} > 0$, $M > 0$, $\alpha \in (0, 1)$, $\gamma > 1$, and $x^0 \in \Omega$ be given. Initialize $k \leftarrow 0$.

Step 1. Initialize $j \leftarrow 0$ and $\sigma_{k,j} = \sigma_{\min}$.

Step 2. Choose a symmetric positive semidefinite matrix $B_{k,j} \in \mathbb{R}^{n \times n}$ with $\|B_{k,j}\| \leq M$ and compute $x_{\text{trial}}^{k,j}$ as the global minimizer of

$$\text{Minimize}_{x \in \Omega} \Psi(x; x^k, B_{k,j}, \delta, \sigma_{k,j}). \quad (6) \quad \{\text{ovo-subpro}\}$$

Step 3. If

$$f(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 \quad (7) \quad \{\text{ovo-armijo}\}$$

does not hold, set $\sigma_{k,j+1} = \gamma \sigma_{k,j}$, $j \leftarrow j + 1$, and go to Step 2.

Step 4. Set $x^{k+1} = x_{\text{trial}}^{k,j}$, $\sigma_k = \sigma_{k,j}$, $j_k = j$, $k \leftarrow k + 1$, and go to Step 1.

The bound $\|B_{k,j}\| \leq M$ keeps the curvature from dominating the model. The freedom in choosing $B_{k,j}$ lets the method accommodate standard quasi-Newton updates such as BFGS or SR1, and the choice $B_{k,j} = 0$ for all k, j recovers the first-order method of [21].

The requirements on $x_{\text{trial}}^{k,j}$ are mild. The theory only needs $x_{\text{trial}}^{k,j}$ to be a stationary point of (6) with $\Psi(x_{\text{trial}}^{k,j}; x^k, B_{k,j}, \delta, \sigma_{k,j}) \leq 0$; since $\Psi(x^k; x^k, B_{k,j}, \delta, \sigma_{k,j}) = 0$, any descent method started at x^k produces such iterates. Moreover, (6) minimizes a strongly convex function over the compact convex set Ω , so it has a unique global minimizer and every stationary point is global; standard convex solvers apply directly.

3 Optimality condition and complexity

Assumption A1. The functions $f_1, \dots, f_m$ belong to $\mathcal{C}^1(\Omega)$, that is, they are continuously differentiable on Ω . {a1}

We begin by establishing that local minimizers of (3) are also minimizers of a related subproblem involving the model Ψ . This result is fundamental because it connects the geometry of the original nonsmooth problem with the smooth subproblems solved at each iteration, and it will be used to derive the optimality condition for (3).

Theorem 3.1. *Suppose that Assumption A1 is satisfied. If x^* is a local minimizer of problem (3), then it is also a local minimizer of the problem*

$$\text{Minimize}_{x \in \Omega} \Psi(x; x^*, 0, 0). \quad (8)$$

Proof. Suppose for contradiction that x^* is not a local minimizer of (8). Then there exists $y^* \in \Omega$ such that $\Psi(y^*; x^*, 0, 0) < \Psi(x^*; x^*, 0, 0) = 0$, which means that for all $i \in I_0(x^*)$,

$$\nabla f_i(x^*)^T(y^* - x^*) < 0.$$

Note that the curvature term vanishes here since $\sigma = 0$ and $B = 0$ in (8). By Assumption A1, each f_i is differentiable on Ω , so

$$\lim_{t \rightarrow 0} \frac{f_i(x^* + t(y^* - x^*)) - f_i(x^*)}{t} = \nabla f_i(x^*)^T(y^* - x^*) < 0, \quad \forall i \in I_0(x^*).$$

Therefore, there exists $\bar{t}_i > 0$ such that $f_i(x^* + t(y^* - x^*)) < f_i(x^*)$ for all $t \in (0, \bar{t}_i]$. Setting $\bar{t} = \min_{i \in I_0(x^*)} \{\bar{t}_i\}$, we obtain

$$f_i(x^* + t(y^* - x^*)) < f_i(x^*) = f(x^*) \quad \text{for all } i \in I_0(x^*) \text{ and } t \in (0, \bar{t}]. \quad (9)$$

For indices $j \notin I_0(x^*)$, the continuity of $f_1, \dots, f_m$ guarantees that, for t sufficiently small,

$$f_i(x^* + t(y^* - x^*)) < f_j(x^* + t(y^* - x^*)) \quad \text{whenever } i \in I(x^*) \text{ and } f(x^*) < f_j(x^*), \quad (10)$$

$$f_i(x^* + t(y^* - x^*)) > f_j(x^* + t(y^* - x^*)) \quad \text{whenever } i \in I(x^*) \text{ and } f(x^*) > f_j(x^*). \quad (11)$$

Inequalities (10)–(11) imply that the relative ordering of the active indices $I_0(x^*)$ within the full ranking of $\{f_i(x^* + t(y^* - x^*))\}_{i=1}^m$ is preserved from x^* to $x^* + t(y^* - x^*)$ for all t small enough. In particular, the p -th position in the ranking is still occupied by some index $i \in I_0(x^*)$, so

$$f_i(x^* + t(y^* - x^*)) = f_{i_p(x^* + t(y^* - x^*))}(x^* + t(y^* - x^*)) = f(x^* + t(y^* - x^*)).$$

Combined with (9), this gives $f(x^* + t(y^* - x^*)) < f(x^*)$ for all t small enough. Since Ω is convex, $x^* + t(y^* - x^*) \in \Omega$ for all $t \in [0, 1]$, contradicting the assumption that x^* is a local minimizer of (3). $\square$

Theorem 3.2. *Suppose that Assumption A1 is satisfied and let $\sigma \geq 0$. If x^* is a local minimizer of problem (3), then it is also a local minimizer of the problem*

$$\text{Minimize}_{x \in \Omega} \Psi(x; x^*, 0, \sigma). \quad (12)$$

164 *Proof.* Suppose for contradiction that x^* is not a local minimizer of (12). Then there exists
 165 $y^* \in \Omega$ such that $\Psi(y^*; x^*, 0, \sigma) < \Psi(x^*; x^*, 0, \sigma) = 0$, which means that

$$166 \quad \max_{i \in I_0(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} + \frac{1}{2} [(y^* - x^*)^T B(y^* - x^*) + \sigma \|y^* - x^*\|^2] < 0.$$

167 Since B is positive semidefinite and $\sigma \geq 0$, the second term satisfies

$$168 \quad \frac{1}{2} [(y^* - x^*)^T B(y^* - x^*) + \sigma \|y^* - x^*\|^2] \geq 0.$$

169 Therefore, we necessarily have

$$170 \quad \max_{i \in I_0(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} < 0,$$

171 which means that $\Psi(y^*; x^*, 0, 0) < 0$. This contradicts Theorem 3.1, which guarantees that
 172 x^* is a minimizer of $\Psi(\cdot; x^*, 0, 0)$ over Ω and hence $\Psi(y^*; x^*, 0, 0) \geq \Psi(x^*; x^*, 0, 0) = 0$ for all
 173 $y^* \in \Omega$. □

174 **Theorem 3.3.** Suppose that Assumption A1 is satisfied and let $\sigma, \delta \geq 0$. If x^* is a local
 175 minimizer of problem (3), then it is also a local minimizer of the problem {teo:conv3}

176

$$\text{Minimize } \Psi(x; x^*, \delta, \sigma). \quad (13) \quad \text{{ovo-problem-2}}$$

177 *Proof.* Suppose for contradiction that x^* is not a local minimizer of (13). Then there exists
 178 $y^* \in \Omega$ such that $\Psi(y^*; x^*, \delta, \sigma) < \Psi(x^*; x^*, \delta, \sigma) = 0$, which means that

$$179 \quad \max_{i \in I_\delta(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} + \frac{1}{2} [(y^* - x^*)^T B(y^* - x^*) + \sigma \|y^* - x^*\|^2] < 0.$$

180 Since the maximum is taken over $I_\delta(x^*) \supseteq I_0(x^*)$, the above inequality implies in particular that
 181 for all $i \in I_\delta(x^*)$,

$$182 \quad \nabla f_i(x^*)^T(y^* - x^*) + \frac{1}{2} [(y^* - x^*)^T B(y^* - x^*) + \sigma \|y^* - x^*\|^2] < 0,$$

183 and since $I_0(x^*) \subseteq I_\delta(x^*)$, this holds in particular for all $i \in I_0(x^*)$. The key observation is that
 184 enlarging the active index set from $I_0(x^*)$ to $I_\delta(x^*)$ can only increase the value of the maximum
 185 in the model, yet the inequality still holds. Therefore,

$$186 \quad \max_{i \in I_0(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} + \frac{1}{2} [(y^* - x^*)^T B(y^* - x^*) + \sigma \|y^* - x^*\|^2] < 0,$$

187 which means $\Psi(y^*; x^*, 0, \sigma) < 0$. This contradicts Theorem 3.2, which establishes that x^* is a
 188 minimizer of $\Psi(\cdot; x^*, 0, \sigma)$ over Ω and hence $\Psi(y^*; x^*, 0, \sigma) \geq 0$ for all $y^* \in \Omega$. □

189 In the spirit of [21], Theorems 3.1–3.4 motivate the following approximate optimality condi-
 190 tion, which will serve as the stopping criterion for Algorithm 2.1. The condition measures how
 191 far a given point is from satisfying the necessary conditions for local optimality of (3), using the
 192 active index set $I_\delta(x)$ to account for the nonsmooth structure of f .

{def:optcond}

193 **Definition 3.1.** Given $\epsilon > 0$ and $\delta > 0$, we say that $x \in \Omega$ satisfies the approximate optimality
 194 condition $C_{\delta,\epsilon}$ if there exist $\mu \in \Sigma_{|I_\delta(x)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that $(\ell_i - x_i)\nu_i^\ell = (x_i - u_i)\nu_i^u = 0$
 195 for all $i = 1, 2, \dots, n$, and

$$196 \quad \left\| \sum_{i \in I(x, \delta)} \mu_i \nabla f_i(x) + \sum_{i=1}^n \left(\nu_i^u - \nu_i^\ell \right) e^i \right\| \leq \epsilon, \quad (14) \quad \{\text{miaprox-box}\}$$

197 where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$, and $\Sigma_q \subset \mathbb{R}^q$ denotes the q -simplex defined
 198 by

$$199 \quad \Sigma_q = \left\{ \lambda \in \mathbb{R}^q \mid \sum_{j=1}^q \lambda_j = 1 \text{ and } \lambda_j \geq 0 \text{ for } j = 1, \dots, q \right\}.$$

200 The condition $C_{\delta,\epsilon}$ can be interpreted as an approximate stationarity requirement: the
 201 weighted combination of gradients of the active component functions, corrected by the mul-
 202 tipliers associated with the box constraints, must be small in norm. When $\epsilon = 0$ and $\delta = 0$, this
 203 reduces to an exact necessary optimality condition for (3), as guaranteed by Theorem 3.4.

{teo:conv3}

204 **Theorem 3.4.** Suppose that Assumption A1 is satisfied. Then, for every k and $j = 0, \dots, j_k$,
 205 there exist $\mu \in \Sigma_{|I(x^k, \delta)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that

$$206 \quad x_{\text{trial}}^{k,j} - x^k = \frac{1}{\sigma_{k,j}} \left[\sum_{i=1}^n \left(\nu_i^\ell - \nu_i^u \right) e^i - \sum_{i \in I(x^k, \delta)} \mu_i \nabla f_i(x^k) \right]. \quad (15) \quad \{\text{eqcoro1}\}$$

207 *Proof.* See []. The result follows from applying the KKT conditions to the subproblem (6),
 208 where the presence of the curvature matrix $B_{k,j}$ does not affect the structure of the optimality
 209 conditions since $B_{k,j}$ enters only through the quadratic term, which vanishes in the gradient
 210 expression evaluated at the minimizer. $\square$

{a2}

211 **Assumption A2.** For all k and $j = 0, \dots, j_k$, the Lagrange multipliers ν^ℓ and ν^u of Theorem 3.4
 212 satisfy $\max_{\{i=1, \dots, n\}} \{|\nu_i^\ell - \nu_i^u|\} \leq c_\nu$ for some constant $c_\nu > 0$ that depends neither on k nor on
 213 j .

214 The Mangasarian-Fromovitz Constraint Qualification (MFCQ) guarantees that, for every
 215 k and $j \in \{0, \dots, j_k\}$, the Lagrange multipliers ν^ℓ and ν^u of Theorem 3.4 are bounded by a
 216 constant (see [15]), which implies that $\max_{\{i=1, \dots, n\}} \{|\nu_i^\ell - \nu_i^u|\}$ is bounded. Assumption A2
 217 strengthens this by requiring a single uniform bound that holds simultaneously for all outer
 218 iterations k and all inner iterations $j \in \{0, \dots, j_k\}$. In the context of the present method, this
 219 is a natural requirement: since the feasible set Ω is compact and the subproblem (6) is solved
 220 over Ω at every iteration, the multipliers ν^ℓ and ν^u are controlled by the geometry of Ω and the
 221 gradient norms of the active functions, both of which remain bounded under Assumption A1.
 222 It is worth noting that the presence of the curvature matrix $B_{k,j}$ in the subproblem does not
 223 affect the structure of the box-constraint multipliers, since $B_{k,j}$ enters only through the interior
 224 quadratic term and does not alter the active constraints at the solution.

{a3}

225 **Assumption A3.** $\|\nabla f_1(x)\|, \|\nabla f_2(x)\|, \dots, \|\nabla f_m(x)\|$ are bounded from above by a constant c_∇
 226 for all $x \in \Omega$.

227 Assumption A3 is standard in the analysis of first- and second-order methods over com-
 228 pact domains and is automatically satisfied when each f_i is continuously differentiable and Ω
 229 is bounded, as is the case here under Assumption A1. Combining Theorem 3.4 with Assump-
 230 tions A2 and A3, we obtain a bound on the step size at each inner iteration. Specifically, from
 231 (15),

$$232 \quad \|x_{\text{trial}}^{k,j} - x^k\| \leq \frac{1}{\sigma_{k,j}} \left[\sum_{i=1}^n \left| \nu_i^\ell - \nu_i^u \right| + c_\nabla \right],$$

233 and applying Assumption A2 yields

$$234 \quad \|x_{\text{trial}}^{k,j} - x^k\| \leq \frac{c_x}{\sigma_{k,j}}, \quad (16) \quad \{\text{ccotasub}\}$$

235 where $c_x = n c_\nu + c_\nabla$. This bound shows that the step length is inversely proportional to the
 236 regularization parameter $\sigma_{k,j}$: as $\sigma_{k,j}$ grows due to repeated rejections in Step 3, the trial point
 237 is forced closer to the current iterate, which is the mechanism by which the algorithm eventually
 238 satisfies the sufficient descent condition (7).

{a4}

239 **Assumption A4.** There exists $L > 0$ such that, for $i = 1, \dots, m$ and all $x, y \in \Omega$,

$$240 \quad f_i(y) \leq f_i(x) + \nabla f_i(x)^T(y - x) + \frac{L}{2} \|y - x\|^2. \quad (17) \quad \{\text{taylor}\}$$

241 Assumption A4 is the standard descent lemma for functions with Lipschitz-continuous gradi-
 242 ents. It implies that the linear model $f_i(x) + \nabla f_i(x)^T(y - x)$ overestimates $f_i(y)$ up to a quadratic
 243 error controlled by L . In the context of the present method, this assumption is used to show
 244 that the model Ψ provides a sufficiently accurate upper bound on the true function f when $\sigma_{k,j}$
 245 is large enough relative to L , which is what ultimately guarantees the finite termination of the
 246 inner loop in Step 3.

{lem1}

247 **Lemma 3.1.** Let $a_1, \dots, a_r \in \mathbb{R}$, $b_1, \dots, b_r \in \mathbb{R}$, $\beta > 0$, and $\{i_1, \dots, i_r\} = \{1, \dots, r\}$ be such
 248 that $a_1 \leq \dots \leq a_r$, $b_j \leq a_j - \beta$ for $j = 1, \dots, r$, and $b_{i_1} \leq \dots \leq b_{i_r}$. Then, $b_{i_q} \leq a_q - \beta$ for
 249 $q = 1, \dots, r$.

250 *Proof.* See [3]. This purely combinatorial result states that if each element of a sequence $\{b_j\}$ lies
 251 strictly below the corresponding element of a sorted sequence $\{a_j\}$ by at least β , then the same
 252 gap is preserved after reordering $\{b_j\}$. In the complexity analysis that follows, this lemma is
 253 applied to transfer descent guarantees from individual component functions f_i to the order-value
 254 function f itself. $\square$

255 The following theorem establishes a sufficient condition on the regularization parameter $\sigma_{k,j}$
 256 that guarantees the acceptance of the trial point $x_{\text{trial}}^{k,j}$ at Step 3 of Algorithm 2.1. It shows that
 257 once $\sigma_{k,j}$ exceeds a threshold determined by the Lipschitz constant L of the gradients and the
 258 tolerance band parameter δ , the sufficient descent condition (7) is necessarily satisfied. This

result is the key ingredient for proving that the inner loop of the algorithm terminates in a finite number of steps, independently of the outer iteration counter k . Notably, the threshold depends on δ^{-1} , which reflects the fact that a narrower active index band requires a stronger regularization to ensure that the trial point remains within the region where the model accurately represents f .

Theorem 3.5. *Suppose that Assumptions A1, A3, and A4 hold. Then, for all k and $j = 0, \dots, j_k$, if*

$$\sigma_{k,j} \geq \max \{L + 2\alpha, 9c_x^2/(2\delta)\}, \quad (18) \quad \{\text{este}\}$$

then $x_{\text{trial}}^{k,j}$ satisfies (7).

Proof. By Assumption A4, there exists $L > 0$ such that, for all $i = 1, 2, \dots, m$,

$$\begin{aligned} f_i(x_{\text{trial}}^{k,j}) &\leq f_i(x^k) + \nabla f_i(x^k)^T (x_{\text{trial}}^{k,j} - x^k) + \frac{L}{2} \|x_{\text{trial}}^{k,j} - x^k\|^2 \\ &\leq f_i(x^k) + \nabla f_i(x^k)^T (x_{\text{trial}}^{k,j} - x^k) + \left(\frac{\sigma}{2} - \alpha\right) \|x_{\text{trial}}^{k,j} - x^k\|^2 \\ &\leq f_i(x^k) + \nabla f_i(x^k)^T (x_{\text{trial}}^{k,j} - x^k) + \frac{1}{2} \left[(x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k) + \sigma \|x_{\text{trial}}^{k,j} - x^k\|^2 \right] \\ &\quad - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 - \frac{1}{2} (x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k). \end{aligned}$$

Since B is a positive semi-definite matrix, it holds that $(x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k) \geq 0$. Therefore, dropping the non-positive term $-\frac{1}{2} (x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k)$, we obtain

$$\begin{aligned} f_i(x_{\text{trial}}^{k,j}) &\leq f_i(x^k) + \nabla f_i(x^k)^T (x_{\text{trial}}^{k,j} - x^k) + \frac{1}{2} \left[(x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k) + \sigma \|x_{\text{trial}}^{k,j} - x^k\|^2 \right] \\ &\quad - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 \end{aligned}$$

Note that $\Psi(x^k; x^k, \delta, \sigma) = 0$. Hence,

$$\nabla f_i(x^k)^T (x_{\text{trial}}^{k,j} - x^k) + \frac{1}{2} \left[(x_{\text{trial}}^{k,j} - x^k)^T B (x_{\text{trial}}^{k,j} - x^k) + \sigma \|x_{\text{trial}}^{k,j} - x^k\|^2 \right] \leq 0,$$

for all $i \in I_\delta(x^k)$. Therefore,

$$f_i(x_{\text{trial}}^{k,j}) \leq f_i(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2, \quad (19) \quad \{\text{cota-lem2}\}$$

for all $i \in I_\delta(x^k)$. On the other hand, by (16) and the Cauchy-Schwarz inequality, if $\sigma_{k,j} \geq \max\{L + 2\alpha, 9c_x^2/(2\delta)\}$, then

$$f_i(x_{\text{trial}}^{k,j}) - \frac{\delta}{3} \leq f_i(x^k) \leq f_i(x_{\text{trial}}^{k,j}) + \frac{\delta}{3}, \quad (20) \quad \{\text{teo7eq1}\}$$

for $i = 1, \dots, m$. This means that the relative ordering of the values $\{f_i(x_{\text{trial}}^{k,j})\}_{i=1}^m$ around the order-value $f(x^k)$ is preserved: specifically, any index i satisfying $f_i(x^k) < f(x^k) - \delta$ also satisfies $f_i(x_{\text{trial}}^{k,j}) < f(x^k)$, and any index i satisfying $f_i(x^k) > f(x^k) + \delta$ also satisfies $f_i(x_{\text{trial}}^{k,j}) > f(x^k)$.

287 Therefore, $i_p(x_{\text{trial}}^{k,j}) \in I_\delta(x^k)$. Now, let $I_\delta(x^k) = \{i_1, \dots, i_r\}$ with $f_{i_1}(x^k) \leq \dots \leq f_{i_r}(x^k)$, and let
 288 $\{i'_1, \dots, i'_r\}$ denote the reordering of these indices such that $f_{i'_1}(x_{\text{trial}}^{k,j}) \leq \dots \leq f_{i'_r}(x_{\text{trial}}^{k,j})$. Since
 289 (19) holds for all $i \in I_\delta(x^k)$, we have $f_{i_j}(x_{\text{trial}}^{k,j}) \leq f_{i_j}(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2$ for $j = 1, \dots, r$. By
 290 the definition of the reordering, for any $q \in \{1, \dots, r\}$,

$$291 \quad f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i'_q}(x_{\text{trial}}^{k,j}) \leq \dots \leq f_{i'_r}(x_{\text{trial}}^{k,j}) \leq f_{i_r}(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2.$$

292 Since $\{i'_q, \dots, i'_r\} \subseteq \{i_1, \dots, i_r\}$ has $r - q + 1$ elements, there exists $\tilde{q} \leq q$ such that $i_{\tilde{q}} \in$
 293 $\{i'_q, \dots, i'_r\}$, and therefore

$$294 \quad f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i_{\tilde{q}}}(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 \leq f_{i_q}(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2.$$

295 Since $i_p(x_{\text{trial}}^{k,j}) \in I_\delta(x^k)$, there exists $q \in \{1, \dots, r\}$ such that $i_p(x_{\text{trial}}^{k,j}) = i'_q$ and $f_{i_q}(x^k) \leq$
 296 $f_{i_p(x^k)}(x^k) = f(x^k)$. Therefore,

$$297 \quad f(x_{\text{trial}}^{k,j}) = f_{i_p(x_{\text{trial}}^{k,j})}(x_{\text{trial}}^{k,j}) = f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i_q}(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2,$$

298 which is precisely (7), completing the proof. $\square$

299 The following theorem summarizes the worst-case complexity of Algorithm 2.1. It establishes
 300 that the inner loop terminates in a finite number of steps at every outer iteration, and provides
 301 explicit bounds on the total number of outer iterations and function evaluations required to
 302 find a point satisfying the approximate optimality condition $C_{\delta,\epsilon}$. The presence of the curvature
 303 matrix $B_{k,j}$ does not affect the order of the complexity bounds, but may reduce the number of
 304 iterations required in practice by producing higher quality steps.

305 **Theorem 3.6.** *Suppose that Assumptions A1, A3, and A4 hold and that there exists $f_{\text{low}} \in \mathbb{R}$*
 306 *such that $f(x) \geq f_{\text{low}}$ for all $x \in \Omega$. Then, σ_k satisfies*

$$307 \quad \sigma_k \leq \gamma \max \{L + 2\alpha, 9c_x^2/(2\delta)\}, \quad (21) \quad \{\text{cota-sigma}\}$$

308 and at most

$$309 \quad \left\lceil 1 + \log_\gamma \left(\frac{\sigma_k}{\sigma_{\min}} \right) \right\rceil \quad (22) \quad \{\text{cota-fcnt}\}$$

310 functional evaluations are performed to obtain (7) at each outer iteration. Moreover, the number
 311 of outer iterations k at which $C_{\delta,\epsilon}$ is not satisfied by x^k is bounded above by

$$312 \quad \left\lceil \frac{\gamma^2 \max \{L + 2\alpha, 9c_x^2/(2\delta)\}^2}{\alpha} \cdot \frac{f(x^0) - f_{\text{low}}}{\epsilon^2} \right\rceil. \quad (23) \quad \{\text{cota-iter}\}$$

313 *Proof.* The bound (21) and the functional evaluation count (22) follow directly from Theorem 18
 314 and the update rule $\sigma_{k,j+1} = \gamma \sigma_{k,j}$ in Step 3: since the initial value at each outer iteration is
 315 $\sigma_{k,0} = \sigma_{\min}$, the inner loop terminates after at most $\lceil 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rceil$ increases before (18) is
 316 satisfied.

For the second part, let $K \subset \mathbb{N}$ be the set of outer iteration indices k at which $C_{\delta,\epsilon}$ is not satisfied by x^{k+1} . For each $k \in K$, since $x^{k+1} = x_{\text{trial}}^{k,j}$ satisfies (7) but does not satisfy $C_{\delta,\epsilon}$, by Theorem 3.4 there exist $\mu \in \Sigma_{|I(x^k,\delta)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that

$$x^{k+1} - x^k = \frac{1}{\sigma_k} \left[\sum_{i=1}^n (\nu_i^\ell - \nu_i^u) e^i - \sum_{i \in I(x^k,\delta)} \mu_i \nabla f_i(x^k) \right],$$

and since $C_{\delta,\epsilon}$ does not hold at x^{k+1} ,

$$\|x^{k+1} - x^k\| = \frac{1}{\sigma_k} \left\| \sum_{i \in I(x^k,\delta)} \mu_i \nabla f_i(x^k) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e^i \right\| > \frac{\epsilon}{\sigma_k}.$$

Using (21), we obtain

$$\|x^{k+1} - x^k\| > \frac{\epsilon}{\gamma \max \{L + 2\alpha, 9c_x^2/(2\delta)\}}.$$

On the other hand, since x^{k+1} satisfies (7),

$$f(x^{k+1}) \leq f(x^k) - \alpha \|x^{k+1} - x^k\|^2 < f(x^k) - \frac{\alpha \epsilon^2}{\gamma^2 \max \{L + 2\alpha, 9c_x^2/(2\delta)\}^2}.$$

Summing over all $k \in K$,

$$\sum_{k \in K} (f(x^k) - f(x^{k+1})) > |K| \cdot \frac{\alpha \epsilon^2}{\gamma^2 \max \{L + 2\alpha, 9c_x^2/(2\delta)\}^2}.$$

Since $f(x) \geq f_{\text{low}}$ for all $x \in \Omega$, the left-hand side is bounded above by $f(x^0) - f_{\text{low}}$, yielding

$$|K| < \frac{\gamma^2 \max \{L + 2\alpha, 9c_x^2/(2\delta)\}^2}{\alpha} \cdot \frac{f(x^0) - f_{\text{low}}}{\epsilon^2},$$

from which (23) follows. $\square$

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